Lecture 10. Monte Carlo Integration

Naive Monte Carlo, importance sampling, antithetic variables, control variables

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10.1 Naive Monte Carlo

To compute $\mu = \mathbb{E}f(X) = \int f(x)p(x)\,dx$ for a density p, draw $X_1, \dots, X_n$ i.i.d. from p and use

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

It is unbiased; by the strong law $\hat{\mu}_n \to \mu$ almost surely when $\mathbb{E}|f(X)| < \infty$; and if $\sigma^2 = \operatorname{Var} f(X) < \infty$, the CLT gives $\sqrt{n}(\hat{\mu}_n - \mu) \Rightarrow N(0, \sigma^2)$. With the sample variance $\hat{\sigma}^2$ of the $f(X_i)$, an approximate 95% interval is $\hat{\mu}_n \pm 1.96\,\hat{\sigma}/\sqrt{n}$.

The error is $O_p(n^{-1/2})$ *regardless of the dimension*, while deterministic quadrature on a grid suffers from the curse of dimensionality. In one or two dimensions, quadrature (Lecture 8) is far more accurate; Monte Carlo wins in high dimensions. To improve on $n^{-1/2}$ one cannot do much, but the constant σ^2 can be reduced: this is **variance reduction**.

Example 10.1

- $\int_0^\infty \sqrt{x}e^{-x}dx = \Gamma(3/2) = \sqrt{\pi}/2$ is $\mathbb{E}\sqrt{X}$ for $X \sim \text{Exp}(1)$; the average of $\sqrt{X_i}$ estimates it with finite variance.
- $\int_0^1 x^{-1/2}dx = 2$ is $\mathbb{E}U^{-1/2}$, but $\mathbb{E}U^{-1} = \infty$: the estimator is consistent yet has infinite variance, so the CLT interval is meaningless and the running averages show occasional large jumps. The substitution $x = y^2$ turns the integral into $\int_0^1 2\,dy$, estimated exactly. A transformation that flattens the integrand is the cheapest variance reduction.
- Buffon's needle: a needle of length L dropped on a floor ruled with parallel lines at spacing $D \geq L$ crosses a line with probability $2L/(\pi D)$. The centre's distance to the nearest line is $Y \sim \text{Unif}(0, D/2)$ and the angle $\Theta \sim \text{Unif}(0, \pi/2)$, independent; the needle crosses iff $Y \leq \frac{L}{2}\sin\Theta$, which has probability $\frac{2}{D}\cdot\frac{2}{\pi}\int_0^{\pi/2}\frac{L}{2}\sin\theta\,d\theta = \frac{2L}{\pi D}$. The crossing frequency thus estimates π.

10.2 Importance sampling

For any density h with $h > 0$ wherever $fp \neq 0$,

$$\mu = \int f(x)\frac{p(x)}{h(x)}h(x)\,dx = \mathbb{E}_h\Big[f(Y)\frac{p(Y)}{h(Y)}\Big], \qquad \hat{\mu}_h = \frac{1}{n}\sum_{i=1}^{n} f(Y_i)w(Y_i), \quad Y_i \overset{\text{iid}}{\sim} h,$$

with **importance weights** $w = p/h$. It is unbiased with variance $\frac{1}{n}\{\int f^2p^2/h - \mu^2\}$. This is useful when p is hard to sample, or when p rarely visits the region where f is large.

Proposition 10.1

The variance of $\hat{\mu}_h$ is minimized by $h^\star \propto |f|\,p$. If $f \geq 0$, then $h^\star = fp/\mu$ and the variance is zero.

Proof. By Cauchy–Schwarz, for every admissible h,

$$\Big(\int |f|p\Big)^2 = \Big(\int \frac{|f|p}{\sqrt{h}}\sqrt{h}\Big)^2 \leq \int \frac{f^2p^2}{h}\int h = \int \frac{f^2p^2}{h},$$

with equality iff $|f|p/\sqrt{h} \propto \sqrt{h}$, i.e. $h \propto |f|p$. If $f \geq 0$, $fw \equiv \mu$ under $h^\star$. ∎

$h^\star$ needs μ itself, so it is unusable, but it is a guide: choose h roughly proportional to $|f|p$, and with tails at least as heavy as fp; otherwise the weights explode and the variance may be infinite. A useful diagnostic is the spread of the weights.

Example 10.2 (Normal tail probability)

Estimate $\pi(c) = P(Z > c) = \Phi(-c)$ for large c. The naive estimator $\frac{1}{n}\sum \mathbf{1}\{Z_i > c\}$ has relative error $\sqrt{(1-\pi(c))/(n\pi(c))}$, which explodes as c grows: almost no draw lands in the tail. Shift the sampling distribution to $h = \phi(\cdot - c)$, i.e. $Y_i \sim N(c, 1)$. The weight is $\phi(y)/\phi(y-c) = e^{-cy+c^2/2}$, so

$$\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Y_i > c\} e^{-cY_i + c^2/2}.$$

Its second moment per draw is

$$e^{c^2}\int_c^\infty e^{-2cy}\phi(y-c)\,dy = e^{c^2}\int_c^\infty \phi(y+c)\,dy = e^{c^2}\Phi(-2c),$$

since $-2cy - \frac{1}{2}(y-c)^2 = -\frac{1}{2}(y+c)^2$. Hence $\operatorname{Var}\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\{e^{c^2}\Phi(-2c) - \pi(c)^2\}$. By Mills' ratio $\Phi(-t) \approx \phi(t)/t$ as $t \to \infty$, the squared relative error is about $c\sqrt{\pi/2}/n$, against $\sqrt{2\pi}\, c e^{c^2/2}/n$ for the naive estimator: the relative error grows like $c^{1/2}$ instead of $c^{1/2}e^{c^2/4}$.

10.3 Antithetic variates

If X and X' have the same distribution but are negatively correlated, then

$$\operatorname{Var}\left[\frac{f(X)+f(X')}{2}\right] = \frac{\operatorname{Var} f(X)}{2}(1+\rho), \qquad \rho = \operatorname{Corr}(f(X), f(X')),$$

which beats two independent draws when $\rho < 0$. For uniforms the natural pair is $(U, 1-U)$; for symmetric distributions about m, $(X, 2m - X)$.

Proposition 10.2

If f and g are both nondecreasing (or both nonincreasing), then $\operatorname{Cov}(f(X), g(X)) \geq 0$ for any random variable X for which the covariance exists. Consequently, for monotone h, $\operatorname{Cov}(h(U), h(1-U)) \leq 0$.

Proof. Let Y be an independent copy of X. Since f and g move in the same direction, $\{f(X) - f(Y)\}\{g(X) - g(Y)\} \geq 0$. Its expectation is

$$2\mathbb{E}f(X)g(X) - 2\mathbb{E}f(X)\mathbb{E}g(X) = 2\operatorname{Cov}(f(X), g(X)) \geq 0.$$

For the consequence, take $f = h$ and $g(u) = -h(1-u)$, which is monotone in the same direction as h. ■

Example 10.3

For $\mu = \mathbb{E}e^U = e - 1$: $\operatorname{Var} e^U = \frac{e^2-1}{2} - (e-1)^2 \approx 0.2420$ and $\operatorname{Cov}(e^U, e^{1-U}) = e - (e-1)^2 \approx -0.2342$. The antithetic average $\frac{1}{2}(e^U + e^{1-U})$ has variance $\frac{1}{2}(0.2420 - 0.2342) \approx 0.0039$, against $0.2420/2 = 0.1210$ for two independent uniforms: a reduction of about 97%.

10.4 Control variates

Suppose $Y = g(X)$ has a known mean μ_Y. For every c, $f(X) - c(Y - \mu_Y)$ is unbiased for μ, with variance

$$\operatorname{Var} f(X) + c^2 \operatorname{Var} Y - 2c\operatorname{Cov}(f(X), Y),$$

minimized at $c^\star = \operatorname{Cov}(f(X), Y)/\operatorname{Var} Y$, where it equals $\operatorname{Var} f(X)\,(1-\rho^2)$, $\rho = \operatorname{Corr}(f(X), Y)$. The more correlated the control, the larger the gain. In practice $c^\star$ is estimated by the least-squares slope of $f(X_i)$ on Y_i (a small bias, negligible for large n).

Example 10.4

For $\mathbb{E}e^U$ with control $Y = U$ ($\mu_Y = \frac{1}{2}$, $\operatorname{Var} U = \frac{1}{12}$): $\operatorname{Cov}(e^U, U) = \mathbb{E}Ue^U - \frac{e-1}{2} = 1 - \frac{e-1}{2} \approx 0.1409$, so $c^\star = 12 \times 0.1409 \approx 1.6903$, and the variance per draw drops from 0.2420 to $0.2420 - 12(0.1409)^2 \approx 0.0039$.